

Notes: Kaplan & Strömberg (RES, 2003)

Financial Contracting Theory Meets the Real World: An Empirical Analysis of Venture Capital Contracts

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1 Big picture

- **Question.** How do real-world venture capital contracts compare with the assumptions and predictions of financial contracting theory?
- **Empirical setting.** The paper studies actual financing contracts between venture capitalists and entrepreneurs. These contracts are especially informative because VC transactions are close to the canonical theoretical setting: an outside investor finances an entrepreneur under asymmetric information, moral hazard, incomplete contracting, and potential hold-up.
- **Sample.** The authors analyze 213 VC financing rounds in 119 portfolio companies provided by 14 VC partnerships, covering investments made between 1987 and 1999.
- **Core empirical finding.** VC contracts separately allocate cash flow rights, board rights, voting rights, liquidation rights, and other control rights. These rights are frequently state contingent and tied to financial performance, non-financial performance, actions, default, securities issuance, and founder employment.
- **Main contracting pattern.** If performance is poor, VCs obtain more control and stronger liquidation rights. As performance improves, founders retain or regain control. If performance is very strong, VCs keep cash flow upside but relinquish most control and liquidation rights.
- **Complementarity.** Cash-flow incentives, board rights, voting rights, liquidation rights, staging, and contingencies are used more as complements than substitutes. Firms with stronger VC control are also more likely to have performance-contingent founder equity and staged funding.
- **Theory connection.** The evidence supports several theoretical traditions at once: principal-agent incentive models, incomplete-contracting/control-rights models, debt and liquidation-right theories, and hold-up theories based on inalienable human capital.
- **Important qualification.** The paper is primarily descriptive and interpretive. It does not estimate a randomized causal effect of contract terms. Its contribution is to measure the contract terms that theory usually abstracts from and to map those terms to theoretical predictions.
- **Bottom line.** Real VC contracts are richer and more multidimensional than most financial-contracting models. Theories do well in broad direction, but the paper documents stylized facts that require models with multiple rights, state contingencies, dynamics, and renegotiation.

2 Incremental contribution of the paper

- **First detailed contract-level test of financial contracting theory.** Prior empirical work had studied venture capital governance or contract features, but this paper provides a systematic, contract-level comparison between real VC contracts and a broad set of financial-contracting theories.
- **Disaggregation of rights.** The paper’s key innovation is to treat cash flow rights, board rights, voting rights, liquidation rights, redemption rights, automatic conversion rights, anti-dilution, vesting, non-competes, and staged funding as separable contract instruments. This is more granular than treating a security simply as “debt,” “equity,” or “convertible preferred.”
- **State-contingent control in real contracts.** The paper documents that board and voting rights can switch based on observable financial and non-financial milestones, not only on default. This directly connects empirical VC contracts to theories of state-contingent control such as Aghion–Bolton and Dewatripont–Tirole.
- **Observable but nonstandard contingencies.** Table 3 shows that contracts make rights contingent on product approvals, FDA milestones, patents, hiring a CEO, completing a business plan, founder employment, and securities offerings. This broadens the empirical meaning of “verifiable signals” in contract theory.

- **Complementarity of contract terms.** The cluster analysis in Table 7 shows that incentives, control, liquidation rights, and staged funding are bundled in systematic ways. This is an incremental contribution because most theories vary one instrument at a time.
- **Dynamic contracting evidence.** Table 8 documents how contract terms evolve across financing rounds. Founder cash flow rights and control generally decline as additional financing rounds occur; VC control and liquidation rights tend to rise. This motivates dynamic theories of VC financing and renegotiation.
- **Theory-building contribution.** The paper does not only test existing theory; it supplies a menu of stylized facts for future models. In particular, it points to models that combine control allocation, cash-flow convexity, liquidation rights, contingent financing, staged investment, founder retention, and security conversion.

3 Data and measurement

3.1 Sample construction

The sample contains 213 venture capital investments in 119 portfolio companies by 14 VC partnerships. The authors asked VCs to provide as many portfolio company investments as they were willing to share. For each company, the VCs provided the legal contracts governing the financing round, ownership information, and, when available, business plans, internal VC analyses, and subsequent performance data.

- 98 of the 213 investments are first VC financing rounds.
- 166 rounds were completed between 1996 and early 1999.
- 79 of the 119 companies received their first VC round between 1996 and early 1999.
- The median round commits \$4.5 million in VC financing, of which a median \$3.6 million is released up front.
- The median number of VCs in a round is four.

Interpretation: The sample is not random; it is based on contracts supplied by 14 VC firms. But the authors provide extensive sample description and compare the firms with Venture Economics data. The sample appears to lean toward larger, more established VC partnerships.

3.2 Aggregation of syndicates

When multiple VCs invest in the same financing round, the paper aggregates all VCs and treats them as one investor group. This is natural because the theory being tested usually models a single outside investor financing an entrepreneur. The founders, employees, non-VC investors, and outsiders are treated separately when possible.

Interpretation: Aggregation simplifies the analysis but also hides potential conflicts inside VC syndicates. The paper’s primary question is the investor–entrepreneur contract, not the internal syndicate contract.

3.3 Cash-flow rights

Cash-flow rights are measured as residual cash flow rights, i.e., fully diluted equity ownership. The paper distinguishes several cases:

- minimum VC ownership if founders meet performance and vesting milestones;
- maximum founder and employee vesting if all vesting conditions are met;

- maximum VC ownership if founders fail to satisfy performance or vesting conditions;
- founder liquidation rights and senior or pari passu claims relative to VC claims.

Interpretation: The same financing round can produce different residual cash-flow allocations depending on performance, vesting, conversion, and liquidation outcomes. This is why the paper emphasizes allocation of rights rather than simply security labels.

3.4 Board rights and voting rights

Board rights record board seats reserved for VCs, founders, and outsiders. Voting rights record votes controlled by the VCs, founders, or neither. The paper distinguishes normal-state rights from adverse-state rights.

- A normal state is the board or voting allocation after the financing closes.
- An adverse state is the allocation if the portfolio company performs poorly or triggers a contractual contingency.
- State-contingent control is measured when control changes across states.

Interpretation: This measurement is crucial for incomplete-contracting theory. The relevant empirical object is not only who controls today, but who controls conditional on poor performance.

3.5 Liquidation, redemption, and conversion rights

Liquidation rights determine who receives proceeds in liquidation or bankruptcy. Redemption/put rights let VCs require the firm to repurchase preferred securities after some maturity. Automatic conversion provisions convert VC preferred claims into common equity once the IPO or sale price exceeds a threshold.

Interpretation: Liquidation and conversion rights create state-contingent payoffs. They make VC claims debt-like in bad states and equity-like in good states, matching the logic of convertible preferred securities.

3.6 Contingencies and milestones

Contracts include contingencies based on:

- financial performance;
- non-financial performance such as product development, FDA approval, patent approvals, or customer milestones;
- actions such as hiring key executives or adopting a new business plan;
- default on dividends or redemption;
- future securities issuance or valuation thresholds;
- the founder remaining employed.

Interpretation: These contingencies demonstrate that many events that theory often treats as noncontractible can be made contractible in VC settings. But contracts remain incomplete because control rights still matter.

3.7 Explanatory variables used in regressions

The paper uses several proxies for uncertainty, agency problems, and information quality:

- **Pre-revenue:** whether the venture has no revenues at financing.
- **Repeat entrepreneur:** whether the founder previously created a firm that went public or was sold to a public company.
- **Months since first VC round:** duration of VC-founder relationship.
- **VC cumulative investment:** total VC financing invested to date, instrumented in some regressions.
- **Return:** performance proxy based on the round price relative to first-round price, coded into high/low quartile dummies.
- **Industry volatility, R&D, size, fixed assets, long-term debt:** industry proxies for uncertainty, signal quality, asset tangibility, and capital structure.

Interpretation: These variables are not random treatments. They are cross-sectional proxies that connect observed contract terms to theoretical determinants of optimal contracts.

4 Empirical specifications and theory mapping

4.1 Descriptive allocation of rights

The basic empirical object is a descriptive distribution:

$$Rights_r = \{CashFlow_r, Board_r, Voting_r, Liquidation_r, Conversion_r, Vesting_r, Staging_r\}$$

for financing round r . The paper reports means, medians, and frequencies for all rounds and first VC rounds.

Interpretation: This descriptive stage establishes the basic facts: rights are separated, contingent, and multidimensional.

4.2 Cross-sectional regressions for cash-flow incentives

Table 4 estimates regressions of founder cash-flow rights and pay-performance sensitivity on venture characteristics:

$$FounderIncentive_r = \alpha + \beta_1 PreRevenue_r + \beta_2 RepeatEntrepreneur_r + \beta_3 MonthsSinceFirstRound_r + \beta_4 VCInvestment_r + \beta_5 Return_r + X'_r \gamma + \varepsilon_r.$$

Interpretation: The theoretical prediction is that stronger asymmetric information and moral hazard require stronger cash-flow incentives. Pre-revenue status and early relationship stages proxy for greater uncertainty.

4.3 Cross-sectional regressions for control rights

Table 5 estimates ordered logits for the degree of VC board control and voting control:

$$DegreeControl_r \in \{0, 1, 2, 3\},$$

where zero means founder control, one means neither party controls, two means VC control in bad states only, and three means VC control in both good and bad states.

Interpretation: The Aghion–Bolton style prediction is that as external financing constraints or conflicts worsen, control should move from entrepreneur control to state-contingent investor control to full investor control.

4.4 Cross-sectional regressions for liquidation rights and staging

Table 6 studies liquidation claims, redemption rights, timing until next financing, and the percentage of funds released up front:

$$\begin{aligned} LiquidationOrStaging_r = & \alpha + \beta_1 PreRevenue_r + \beta_2 RepeatEntrepreneur_r \\ & + \beta_3 MonthsSinceFirstRound_r + \beta_4 Investment_r + X_r' \gamma + \varepsilon_r. \end{aligned}$$

Interpretation: The theoretical motivation comes from debt theories and hold-up theories. Liquidation rights and staging can strengthen incentives, protect investors, and reduce founder hold-up.

4.5 Cluster analysis of contract bundles

Table 7 uses hierarchical cluster analysis to group contracts by 12 major contractual characteristics. The clusters are ordered from maximum VC control to maximum founder control.

Interpretation: This analysis asks whether contractual rights are substitutes or complements. The main finding is complementarity: strong VC cash-flow claims, voting control, board control, liquidation rights, and contingencies tend to occur together.

4.6 Dynamic evolution across rounds

Table 8 compares round 1 financings with and without ex ante staging, later rounds, and within-company changes across consecutive rounds.

Interpretation: This dynamic evidence connects to theories of staged financing and renegotiation. As more VC capital is invested and the firm evolves, founder rights decline and VC rights increase, while some contingencies become less common.

5 Identification strategy

5.1 Nature of evidence

This paper is not a causal treatment-effect paper. Its identification strategy is theoretical mapping plus detailed measurement. It identifies stylized facts about real contracts and compares them to predictions of financial contracting theories.

- The descriptive tables identify how rights are allocated in actual contracts.
- Cross-sectional regressions identify correlations between contract terms and proxies for uncertainty, performance, relationship age, and agency severity.
- Cluster analysis identifies systematic complementarities across contractual provisions.
- Dynamic tables identify how contract terms evolve across rounds.

Interpretation: The evidence is strongest when used to test whether broad theoretical predictions are directionally consistent with observed contracts. It is weaker for causal claims about why a specific contract term was chosen.

5.2 What makes the empirical setting credible for theory testing?

- VC financing resembles the investor–entrepreneur setting in financial contracting models.
- VCs have strong incentives to maximize value and few private benefits of control.
- Startups face severe asymmetric information, moral hazard, and hold-up problems.
- Contracts are detailed, negotiated, and designed by sophisticated parties.
- The paper observes legal contracts directly rather than inferring rights from public securities data.

Interpretation: The empirical setting is unusually close to the theoretical primitives. That is why the paper can speak directly to financial-contracting theory.

5.3 Key limitations

- The sample is not random; it is supplied by 14 VC firms willing to provide contracts.
- The paper aggregates VC syndicates, which may hide conflicts among investors.
- Many regressions are cross-sectional and may reflect unobserved venture quality or VC style.
- Some variables have missing observations, so sample size varies across analyses.
- Contract terms may reflect renegotiation and bargaining power, not only ex ante optimal design.

Interpretation: These limitations do not undermine the descriptive contribution. They matter mainly for interpreting regression coefficients as causal.

6 Tables and interpretations

6.1 Table 1: Summary information

Read:

- The sample contains 213 rounds in 119 portfolio companies by 14 VC firms.
- 98 rounds are first VC financing rounds.
- The median committed financing per round is \$4.5 million; the median amount released up front is \$3.6 million.
- The typical financing round involves multiple VCs, with a median of four VCs.
- The sample is concentrated in information technology/software, telecommunications, health care, and related high-growth industries.
- Many rounds are pre-revenue, and repeat entrepreneurs are present in a minority of cases.

Interpretation: Table 1 frames the dataset. These are high-uncertainty, high-growth companies for which contracts must handle information, incentives, control, and staged financing.

Economic message: The sample is precisely the environment where financial-contracting theory predicts complex contracts should matter.

TABLE 1																	
Summary information																	
A. VC financing round statistics	All	1st	2nd	3rd	4th	5th	6th	7th +									
Number of rounds in sample																	
By all companies	213	98	50	31	17	7	5	5									
By companies going public by 31 December 1999	73	29	18	13	6	2	2	3									
By companies acquired by 31 December 1999	32	13	10	5	2	1	1	0									
By companies private/liquidated by 31 December 1999	108	56	22	13	9	4	2	2									
# Rounds in sample/# rounds reported by 1 May 1999:																	
All rounds completed by 1 May 1999	59%	82%	60%	48%	42%	39%	56%	23%									
By companies going public by 31 December 1999	53%	74%	58%	50%	38%	22%	40%	23%									
By companies acquired by 31 December 1999	70%	76%	83%	56%	40%	50%	100%	—									
By companies private/liquidated by 31 December 1999	62%	89%	54%	43%	45%	57%	67%	22%									
VC financing amounts (\$ millions):																	
Total financing committed,	Mean	6.8	6.8	6.3	7.4	6.4	11.1	3.3	8.6								
	Median	4.5	3.6	5.4	6.2	3.0	7.5	2.0	11.1								
Financing provided up front,	Mean	5.2	3.8	5.7	6.9	6.4	9.0	3.3	8.1								
	Median	3.6	3.0	4.7	6.2	3.0	6.6	2.0	9.4								
Number of VCs in syndicate:	Mean	4.3	2.7	4.8	6.4	5.6	6.7	6.6	10.2								
	Median	4.0	2.0	4.5	6.0	5.0	7.0	5.0	9.0								
B. Year of VC financing	Pre92	92	93	94	95	96	97	98	99	Total							
Number of rounds in sample	13	2	3	12	17	53	53	57	3	213							
Year company first received VC financing	10	2	1	13	14	33	21	23	2	119							
C. VC firm characteristics	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Total		
# Rounds	19	47	3	42	16	6	4	2	22	5	13	14	19	1	213		
# Portfolio comp.	9	20	3	22	9	6	4	2	10	4	11	10	8	1	119		
Main location	CA	MW	NE	MW	CA	NE	MW	CA	DIV	NE	MW	DIV	DIV	CA			
															Mean Med		
VC is younger than (yrs)	5	15	20	20	5	5	10	15	25	5	20	30	10	20	12.6 11.0		
VC size rank in terms of managed funds, is top	100	100	25	250	250	—	—	250	25	250	250	25	100	100	123.2 90.5		
D. Industry distribution of companies																	
	Biotech/Medical				IT/Software				Telecom			Healthcare			Retail		Other
Companies	18				50				15			11			11		14
Financing rounds	33				86				21			15			23		35
E. Geographical distribution of companies																	
Companies										California		Midwest		North-east	Other		
										35		25		27	32		
Financing rounds										59		40		60	64		
F. Securities																	
Convertible preferred stock										Financing rounds		Percentage of rounds					
Convertible zero-coupon debt										170		79.8%					
Convertible preferred stock and convertible zero-coupon debt										4		1.9					

(a) Table 1: Summary information, part 1.

<i>G. Other characteristics of the sample</i>	All rounds			First observed round for each company		
	Mean	Median	<i>N</i>	Mean	Median	<i>N</i>
Pre-revenue rounds, %	38.5	—	208	45.7	—	116
Repeat entrepreneur, %	16.9	—	213	16.0	—	119
Volatility of ind. log returns	6.05	5.94	213	6.13	5.99	119
Industry R&D/sales	0.063	0.090	212	0.063	0.090	118
Log industry sales (\$M)	10.0	10.3	205	10.1	10.4	115
Industry fixed/total assets	0.273	0.260	213	0.277	0.270	119
Ind. long-term debt/assets	0.046	0.00	212	0.049	0.00	118

Summary information for 213 investments in 119 portfolio companies by 14 venture capital partnerships. Investments were made between December 1986 and April 1999. Reported rounds are the number of rounds completed before 1 May 1999, based on data from VE combined with contract information. Total financing committed is the amount of equity financing committed to by the VCs in each round. Financing provided up front is the amount of equity financing actually provided to the portfolio company at the round closing. The main location of the VC firms is either California (CA), Midwest U.S. (MW), North-east U.S. (NE), or diverse (DIV), i.e. the VC has several offices and no clear main location. The VC rank is in terms of funds managed in 1999 and was obtained from VE. Pre-revenue rounds are rounds where the venture has no revenues at the time of the closing. Repeat entrepreneurs are founders whose previous ventures were able to go public or were sold to a public company. Industry volatility is the value-weighted volatility of the log stock return for the venture's industry according to the Fama and French (1997) industry classification. Industry R&D/sales is the aggregate R&D expense to sales for public firms in the venture's three-digit SIC industry according to COMPUSTAT. Industry log sales, in millions, are for the venture's four-digit SIC industry according to the 1997 U.S. census. Long-term debt to assets is the median ratio of long-term debt to assets for public firms in the venture's three-digit SIC industry according to COMPUSTAT.

(b) Table 1: Summary information, part 2.

6.2 Table 2: Distribution of cash flow, control, and liquidation rights

Read:

- VCs hold large residual cash-flow stakes, while founders retain meaningful but contingent equity.
- Founder equity is lower in the maximum-VC-ownership state than in the best case.
- Boards average about six members, and VCs often have substantial board representation.
- Board and voting control can change in adverse states.
- Liquidation rights, redemption rights, automatic conversion, anti-dilution, founder vesting, and non-compete clauses are common.

Interpretation: Table 2 is the paper’s core descriptive table. It shows that the same security instrument can embed multiple rights and contingencies. It also demonstrates that VC control is state contingent, not merely fixed at the time of financing.

Economic message: VC contracts look like state-contingent bundles of debt, equity, and governance rights, not like simple securities.

A. Residual cash flow rights													
	Minimum VC ownership contingency		Max. founder and employee vesting contingency		Maximum VC ownership		Difference, min. and max. VC ownership						
	Mean	Median	Mean	Median	Mean	Median	Mean	Median					
All rounds, N = 212													
VC%	46.7	47.3	47.9	47.9	55.5	57.5	-8.8	-4.2					
Founders%	31.1	29.8	30.0	27.1	24.3	20.1	6.8	0.0					
Others%	22.2	20.4	22.1	20.4	20.2	17.2	2.0	0.0					
First VC rounds only, N = 98													
VC%	40.4	41.0***	42.4	43.1***	53.0	50.5*	-12.6	-8.0***					
Founders%	39.5	38.7***	37.7	37.6***	29.6	29.5***	9.9	3.7***					
Others%	20.31	18.8**	19.9	18.8**	17.4	13.2***	2.7	0.0					
B. Board rights													
Mean (median)	All rounds, N = 201				First VC rounds, N = 95								
	Normal		Adverse state		Normal		Adverse state						
Number of board seats	6.0 (6.0)		6.3 (6.0)		5.7 (5.0)***		6.0 (5.0)***						
% VC seats	41.4 (40.0)		46.0 (42.9)		37.0 (40.0)***		42.6 (40.0)***						
% Founder seats	35.4 (37.5)		32.9 (33.3)		38.5 (40.0)***		35.4 (40.0)*						
% Outsider seats	23.2 (20.0)		21.0 (20.0)		24.5 (20.0)		22.0 (20.0)						
% VC board majority	25.4		35.8		11.6***		27.4**						
% Founder majority	13.9		12.4		20.0**		16.8*						
% Neither board majority	60.7		51.7		68.4**		55.8						
% of cases with adverse state board provisions	18.4				21.1								
% Seats to cash flow rights, VC	1.00 (0.89)				1.09 (0.92)								
% Seats to cash flow rights, founder	1.77 (1.12)				1.27 (0.99)								
Signed rank test Z-stat [P-value] VC vs. F	-5.50***				-0.59								
C. Voting rights													
Mean (median)	All rounds, N = 212				First VC rounds, N = 98								
	Minimum VC votes	Maximum VC votes	Difference min. - max.		Minimum VC votes	Maximum VC votes	Difference min. - max.						
% VC votes	53.6 (52.9)	62.3 (64.3)	-8.7 (-1.6)		46.3 (45.4)***	58.9 (59.6)*	-12.6 (-6.4)**						
% Founder votes	33.7 (31.1)	24.5 (20.1)	9.2 (1.5)		42.9 (42.5)***	29.8 (29.5)**	13.0 (6.2)***						
% Others votes	12.6 (7.1)	13.1 (9.1)	-0.5 (0.0)		10.9 (5.4)**	11.3 (5.5)***	-0.4 (0.0)						
% VC control	52.8	68.9			40.8***	61.2**							
% Founder control	23.6	12.3			37.8***	21.4***							
% Neither control	23.6	18.9			21.4	17.3							
% Switch in control	17.8				24.5**								
% Votes to cash flow:													
VC	1.16 (1.13)	1.15 (1.09)			1.16 (1.13)	1.16 (1.07)**							
Founder	1.08 (1.09)	1.02 (1.04)			1.06 (1.08)	0.98 (1.00)*							
Others	0.49 (0.49)	0.61 (0.64)			0.47 (0.43)	0.66 (0.72)							
Signed rank Z, VC vs. F	4.26***	5.01***			3.14***	3.00***							
D. Liquidation rights and redemption rights													

- Contingencies affect cash flow rights, voting control, board control, liquidation rights, redemption rights, rights to force a sale, and release of committed financing.

Interpretation: Table 3 shows that many states that are abstractly “nonverifiable” in theory are in practice made contractible through specific legal milestones. Yet the existence of broad control rights shows that contracts remain incomplete.

Economic message: Real VC contracts combine explicit milestone contracting with residual control allocation.

TABLE 3
Contingencies and the contracting space

A. Percentage of rounds with contingent contracts

	Cash flow rights	Voting rights	Board rights	Right to force sale	Liquidation or dividend amount	Redempt. rights	Release of committed funds	Any rights
Contingent on financial performance	8.0% [11.2]*	5.6% [7.1]	0.5% [1.0]	1.4% [2.0]	3.3% [2.0]	0.5% [1.0]	4.2% [7.1]*	17.4% [22.4]*
Contingent on non-financial performance	6.1 [6.1]	4.2 [3.1]	0.0 [0.0]	0.0 [0.0]	0.0 [0.0]	0.0 [0.0]	3.3 [5.1]	8.9 [11.2]
Contingent on actions	1.9 [2.0]	0.5 [0.0]	0.5 [1.0]	0.0 [0.0]	0.0 [0.0]	0.5 [0.0]	8.9 [12.2]	11.3 [14.3]
Contingent on default on dividend or redemption payment	5.2 [6.1]	5.2 [6.1]	19.2 [21.4]	1.4 [2.0]	2.8 [1.0]	0.5 [0.0]	0.0 [0.0]	27.7 [29.6]
Contingent on future securities offerings or “fair market value”	8.9 [13.3]**	7.5 [9.2]	0.0 [0.0]	3.8 [5.1]	11.7 [16.3]*	0.9 [1.0]	0.9 [1.0]	21.1 [27.6]**
Any of the above	18.8 [23.5]*	15.5 [16.3]	19.2 [22.4]	6.1 [8.2]	15.5 [19.4]	2.3 [2.0]	14.6 [20.4]**	52.6 [60.2]**
Contingent on founder remaining with firm	44.6 [51.0]*	40.4 [48.0]**	—	0.0 [0.0]	0.0 [0.0]	0.0 [0.0]	0.9 [1.0]	40.8 [51.0]*
Any contingent contracting	52.6 [61.2]**	50.7 [58.2]**	19.2 [22.4]	6.1 [8.2]	15.5 [19.4]	2.3 [2.0]	14.6 [20.4]**	72.8 [79.6]**

Contingencies for 213 investments in 119 portfolio companies by 14 venture capital partnerships. Investments were made between 1987 and 1999. Table measures the extent to which venture capital contracts include contingencies based on observable and verifiable characteristics or states. The numbers in the table are the percentage of all rounds that exhibit a certain type of contingent contracting. The numbers in brackets are the corresponding percentages of first VC financings only. The “Contingent on future securities offerings or ‘fair market value’” group excludes anti-dilution and automatic conversion provisions. First VC rounds differ from subsequent rounds at the 1% ***, 5% **, and 10% * levels.

B. Examples of contingencies

Type of contingency	Examples
1. Contract contingent on financial measures of performance	- VC dividend on preferred shares, payable in common stock, is suspended if revenue and operating profit goals attained (CF, VO) - Founder (VC) receives options if revenue/earnings goal (not) attained (CF) - VC can only vote for all owned shares if realized EBIT below threshold value, in which case VC gets voting control (VO) - If net worth below threshold, VC will get three more board seats (BD) - Founder has veto power against sale of company for 3 years, as long as company achieves at least 80% of revenues and profits before tax relative to business plan (FS) - VC liquidation value multiple of net operating cash flow or net earnings (LI) - If earnings are below threshold, VC is allowed to redeem shares (RD) - Committed funds paid out when achieves projected net revenue (ST)
2. Contract contingent on non-financial measures of performance	- Employee shares vest when release of second major version of the product that incorporates significant new functionality, FDA approval of new drug, new corporate partnership found, or patents approved (CF, VO) - VC gets fewer shares if secures distribution agreement within 3 months (CF, VO) - Founder gets options when company secures threshold number of customers who have purchased the product and give positive feedback (CF) - Committed funding paid out when new clinical tests completed, new strategic partnership completed, or patent approved (ST)
3. Contract contingent on	If company cannot pay dividend in cash, it has to be paid in common stock (CF,

(a) Table 3: contingencies, part 1.

TABLE 3—Continued

Type of contingency	Examples
	- If company fails to pay out certain fraction of revenue as a dividend, VC elects majority of board (BD) - If company fails to redeem preferred stock, VC has the right to select an investment banker to assist in the sale of the business (FS) - If company fails to redeem preferred stock, cumulative dividend will be increased (LI)
4. Contract contingent on certain actions being taken	- Founder gets equity when VP of sales and marketing hired, or acquires certain technology (CF, VO) - VC gets board majority if company fails to recruit a CEO within 9 months (BD) - VC has option to redeem preferred stock until new CEO hired acceptable to VC (RD) - Committed funding paid out subject to new business plan for entering new markets completed and approved by board, new key executives or CEO hired, certain company acquired, or company starts to develop new facilities (ST)
5. Contract contingent on offering of securities	- Founder ownership increasing non-linear function of share price obtained in sale or IPO (CF) - VC receives dividend in stock, cancelled upon IPO of minimum value, or if company manages to raise minimum amount of new funding above minimum price per share (CF, VO) - VC warrants expire if company manages to raise alternative funds where proceeds and price of securities exceed threshold (CF) - If no IPO has occurred within 5 years, VC has right to force a sale of company (FS) - VC can redeem preferred shares after IPO at a value that increases the later the IPO date (LI) - VC can redeem shares at fair market value (LI)
6. Contract contingent on founder staying with firm	- Founder loses unvested shares if employment terminated (CF, VO) - Founder loses voting rights (but keeps cash flow rights) for shares if terminated for cause (VO) - Committed funding paid out under the condition that founder still employed (ST)

Contingencies for 213 investments in 119 portfolio companies by 14 venture capital partnerships. Investments were made between 1987 and 1999. Table gives examples of contingencies based on observable and verifiable characteristics or states. Examples include contingent residual cash flow rights (CF), voting majority (VO), board control (BD), right to force a sale (FS), liquidation or dividend cash flow rights (LI), redemption rights (RD), and release of committed financing (ST).

(b) Table 3: contingencies, part 2.

6.4 Table 4: Determinants of founder cash flow incentives

Read:

- Founder residual equity declines as the VC–founder relationship progresses.
- Explicit performance-based equity sensitivity is higher in pre-revenue ventures.
- Performance-benchmark sensitivity is lower for repeat entrepreneurs.
- Time-vesting sensitivity is also higher in pre-revenue ventures.
- Industry characteristics suggest that when explicit signals are noisier, contracts substitute time vesting for explicit performance benchmarks.

Interpretation: Table 4 connects observed contracts to principal-agent theory. Greater uncertainty and asymmetric information are associated with stronger pay-performance sensitivity and more vesting.

Economic message: Cash-flow rights are used as incentives, and contract form changes with the quality of performance signals.

TABLE 4
Determinants of founder cash flow incentives

	% Founder residual cash flow rights (FRCF), best case	% FRCF sensitivity, performance benchmarks	Relative % FRCF sensitivity, performance benchmarks	% FRCF sensitivity, vesting	Relative % FRCF sensitivity, vesting	Founder has liquidation CF rights (logit)
All, $N = 189$	All, $N = 189$	All, $N = 188$	All, $N = 189$	All, $N = 188$	All, $N = 188$	All, $N = 188$
multiple	—	—	—	—	—	—
mutant	—	—	—	—	—	—
revenue	-3.68 (3.17)	1.61 (0.63)**	4.74 (1.69)***	8.47 (2.33)***	23.62 (6.04)***	0.06 (0.56)
repeat entrepr.	6.14 (3.76)	-0.67 (0.47)	-2.20 (1.24)*	-0.06 (2.30)	1.96 (7.09)	1.78 (0.62)*
months since 1st VC round	-0.70 (0.12)**	-0.06 (0.03)**	-0.13 (0.06)**	-0.063 (0.06)	-0.37 (0.20)*	0.028 (0.01)
VC cum. invest (instrumented)	3.67 (2.79)	1.13 (0.67)*	3.20 (1.72)*	-0.37 (1.76)	3.51 (5.38)	—
return	-1.35 (2.44)	-0.81 (0.36)**	-2.34 (1.02)**	0.08 (1.01)	1.86 (5.36)	0.25 (0.44)
volatility	-2.57 (2.36)	-0.45 (0.38)	-2.03 (0.95)**	4.16 (1.60)**	10.76 (4.19)**	0.012 (0.38)
R&D sales	35.0 (37.6)	-20.1 (11.2)*	-69.4 (31.4)**	23.4 (21.7)	17.48 (67.66)	-6.74 (5.76)
industry size	1.53 (1.20)	0.68 (0.28)**	2.43 (0.81)***	-0.72 (0.75)	-0.34 (2.42)	-0.094 (0.20)
leverage	1.86 (3.55)	-2.02 (0.70)**	-6.61 (1.91)***	-1.82 (2.36)	-19.01 (7.01)***	-0.99 (0.58)*
industry	12.9 (4.30)***	-0.44 (0.89)	-0.55 (2.17)	-2.40 (2.06)	-20.91 (6.89)***	-0.67 (0.58)
exit-cost	4.67 (4.00)	-1.31 (0.93)	-4.78 (2.29)**	-1.42 (2.09)	-10.16 (7.13)	-1.63 (0.60)*
1/F-test, location [P-value]	3.70 (0.03)**	6.18 (0.002)***	6.23 (0.002)***	0.08 (0.92)	0.95 (0.38)	2.49 (0.29)
VC dummies	Yes	Yes	Yes	Yes	Yes	Yes
1/F-test, VC [P-value]	2.30 (0.02)**	2.72 (0.008)***	2.94 (0.004)***	1.67 (0.11)	0.84 (0.57)	8.94 (0.18)
Adj. R ²	0.31	0.33	0.40	0.27	0.30	0.22

S and two-stage least squares regressions of founder cash flow incentives on various independent variables for 213 investments in 119 portfolio companies by 14 VC firms. All investments were made between 1987 and 1999. % Founder residual cash flow rights (FRCF), best case is the % of cash flow rights owned by the founder. % FRCF sensitivity, performance (time vesting) benchmarks is the difference in founder's equity if they receive (time vesting) benchmarks. Relative % FRCF sensitivity, performance (time vesting) benchmarks are the corresponding measures as a percentage of the founder's equity stake in the best case. Pre-revenue takes the value of one if the venture has no revenues at the time of financing, and zero otherwise. Repeat entrepr. takes the value of one if the founder's previous venture was able to go public or was sold to a public company. Industry volatility is the value-weighted volatility of the log return for the venture's industry according to the Fama and French (1997) industry classification. Industry R&D sales is the aggregate R&D expense to sales for public in the venture's three-digit SIC industry according to COMPUSTAT. Industry size is equal to log sales, in millions, in the venture's four-digit SIC industry according to 1997 U.S. census. VC cumulative investment is instrumented using two-stage least squares with industry capital expenditures to sales as well as year dummies. Return is the value of return, one if the ratio of post share round price to first round price is in the highest quartile, one if it is in the highest quartile, and zero otherwise. R^2 is the value of adjusted R^2 for the entire sample.

Table 4: Determinants of founder cash flow incentives.

TABLE 5
Determinants of control allocations

	Degree of VC board control (ordered logit)	Degree of VC voting control (ordered logit)	Ln auto. conv. price/round price (2SLS)
Observations	All rounds $N = 182$	All rounds $N = 192$	All rounds $N = 158$
Constant	—	—	—
Pre-revenue	0.81 (0.39)**	1.27 (0.43)***	0.24 (0.10)**
Repeat entrepr.	0.07 (0.50)	-0.83 (0.47)*	-0.25 (0.12)**
Months since 1st VC round	0.066 (0.012)***	0.066 (0.014)***	-0.005 (0.005)
VC cum. invest (instrumented)	—	—	-0.084 (0.088)
Return	0.58 (0.35)*	0.24 (0.32)	-0.36 (0.12)***
Ind. volatility	0.63 (0.29)**	0.87 (0.35)**	0.071 (0.074)
Ind. fixed assets	6.25 (2.51)**	5.54 (3.18)*	-0.67 (0.81)
Industry size	-0.08 (0.17)	-0.29 (0.19)	-0.090 (0.049)*
California	-0.45 (0.48)	-0.74 (0.52)	-0.24 (0.13)*
Midwest	0.44 (0.49)	-1.36 (0.53)**	-0.27 (0.12)**
North-east	1.14 (0.47)**	-1.35 (0.50)**	-0.20 (0.14)
1/F-test, location	$\chi^2(2) = 9.6^{***}$	$\chi^2(2) = 1.8$	$F(2, 138) = 0.12$
[P-value]	[0.01]	[0.42]	[0.89]
VC dummies?	Yes	Yes	Yes
1/F-test, VC	$\chi^2(7) = 20.1^{***}$	$\chi^2(7) = 11.8^*$	$F(8, 138) = 0.27$
[P-value]	[0.01]	[0.11]	[0.27]
Adj. R ²	0.17	0.17	0.57
Ord. logit cut-offs:			
Cut-off group 1	2.91	1.19	
Cut-off group 2	6.12	2.72	
Cut-off group 3	6.90	3.53	

Two-stage least squares and ordered logit regressions of VC board control, voting control, and automatic conversion provisions on various independent variables for 213 investments in 119 portfolio companies by 14 VC partnerships. Investments were made between 1987 and 1999. The degree of board and voting control variables take the value of 0 if the founder always has control, 1 if neither the VC nor the founder has control, 2 if the VCs have control only in the bad state, and 3 if the VCs always have control. Pre-revenue takes the value of one if the venture has no revenues at the time of financing, and zero otherwise. Repeat entrepr. takes the value of one if the founder's previous venture was able to go public or was sold to a public company. Industry volatility is the value-weighted volatility of the log stock return for the firm's industry according to the Fama and French (1997) industry classification. Industry fixed assets is the median fixed to total assets for all (public as well as closely held) firms in the venture's four-digit SIC industry according to Onsource. Industry size is equal to log sales, in millions, in the venture's four-digit SIC industry according to the 1997 U.S. census. VC cumulative investment is instrumented using two-stage least squares with industry capital expenditures to sales as well as year dummies. Return takes the value of minus one if the ratio of post share round price to first round price is in the lowest quartile, one if it is in the highest quartile, and zero otherwise. White (1980) robust standard errors are in parentheses. Asterisks indicate statistical significance at the 1%***, 5%**, and 10%* levels.

Table 5: Determinants of control allocations.

6.5 Table 5: Determinants of control allocations

Read:

- VC board and voting control are more likely in pre-revenue ventures.
- VC control increases with months since the first VC round.
- VC control is more common in high-volatility and fixed-asset-intensive industries.
- Repeat entrepreneurs are associated with lower VC voting control and lower automatic conversion thresholds.
- The automatic conversion price is higher for pre-revenue ventures, meaning VCs require stronger performance before surrendering control.

Interpretation: The table supports incomplete-contracting/control-rights theories. When uncertainty or conflict is high, the investor is allocated more control, especially in bad states.

Economic message: Control rights are not incidental clauses; they are central economic instruments used to govern startups.

6.6 Table 6: Determinants of VC redemption rights, staging, and liquidation claims

Read:

- The table studies whether VC liquidation claims exceed cumulative investment, whether the VC has redemption rights, time until next financing, and fraction of committed funds released up front.
- Repeat entrepreneurs and relationship timing affect liquidation and staging-related terms.
- Some proxies for venture uncertainty and industry characteristics affect the structure of financing release.
- Staging and redemption are important complements to control and cash flow rights.

Interpretation: Table 6 connects VC contracts to debt, liquidation, and staging theories. Liquidation rights protect investors in bad states, while staging lets VCs preserve the option to stop financing if progress is poor.

Economic message: VC financing is not only about how much equity is sold; it is about when funds are released and how downside rights are allocated.

TABLE 6
Determinants of VC redemption rights, staging, and liquidation claims

	VC liq. claims > cum. invest. (logit)	VC has redemption rights (logit)	Months until next fin. round (2SLS)	Months until next fin. round (OLS)	% of funds up front (2SLS)	% of funds up front (2SLS)	% of f (t)
ons	1st firm observ. $N = 108$	1st firm observ. $N = 106$	All observ. $N = 164$	1st firm obs. $N = 98$	All observ. $N = 191$	All observ. $N = 191$	1st firm observ. $N = 108$
entrepreneur	-0.20 (0.75)	-0.32 (0.63)	-0.10 (1.0)	14.7 (7.8)*	-0.30 (3.87)	103.9 (20.8)***	92.6 (4.0)
1st VC round	0.28 (0.98)	-0.83 (0.81)	4.39 (1.86)**	-1.81 (1.23)	3.38 (3.69)	-3.69 (3.62)	-4.30 (9.83)
time in round (instr.)	—	—	-0.003 (0.034)	—	0.21 (0.08)**	0.28 (0.11)**	—
liability	-0.06 (0.52)	-0.20 (0.62)	1.86 (1.45)	—	-3.37 (3.33)	-4.24 (4.22)	-4.56 (9.83)
assets	-3.75 (4.94)	-3.80 (5.08)	5.87 (9.88)	-1.39 (0.86)	-1.72 (2.72)	-3.65 (2.83)	-4.56 (9.83)
ratio to assets	11.84 (5.29)**	5.14 (4.33)	-20.8 (7.1)***	-10.7 (4.9)**	-68.2 (32.4)**	-69.2 (30.9)**	-107.2 (6.65)
size	-0.33 (0.31)	-0.20 (0.35)	0.91 (0.57)	0.62 (0.66)	1.47 (1.75)	2.59 (2.02)	0.65 (0.65)
industry	-2.41 (1.06)**	-1.93 (0.91)**	1.50 (1.43)	—	7.22 (6.39)	—	—
location	-1.04 (0.87)	-0.50 (1.21)	-0.90 (1.61)	—	4.67 (6.16)	—	—
VC dummies	-3.41 (1.18)**	-2.66 (0.92)**	1.77 (1.78)	—	5.56 (5.01)	—	—
R2	0.25	0.23	0.19	0.10	0.39	0.29	0.29

Table 6: Determinants of VC redemption rights, staging, and liquidation claims.

TABLE 7
Complementarity and substitutability of contractual provisions

Cluster:	1—max. VC control	2	3—no party in control	4	5—max. founder control	Kruskal-Wallis χ^2
	$N = 14$	$N = 29$	$N = 17$	$N = 20$	$N = 16$	statistic (4 df)
Founder residual cash flow rights, %:						
Total equity in best case	24.4 (24.0)	25.8 (24.9)	44.6 (45.5)	46.3 (42.1)	52.0 (55.1)	39.2***
Sensitivity, benchmarks	4.54 (0.00)	1.12 (0.00)	0.90 (0.00)	0.36 (0.00)	2.49 (0.00)	12.2***
Sensitivity, vesting	5.61 (4.64)	6.36 (2.35)	23.6 (25.0)	-2.23 (0.00)	-2.67 (-0.69)	40.5***
Voting control (% of cases):						
VC always in control	92.9	89.7	0.0	5.0	12.5	69.6***
Founder always in control	0.0	0.0	0.0	60.0	50.0	42.2***
Control state-contingent	0.0	3.5	100.0	0.0	18.8	75.3***
VC votes/CF, best case	1.14 (1.12)	1.14 (1.12)	1.15 (1.12)	1.18 (1.16)	1.17 (1.17)	1.6
Founder votes/CF, best case	1.12 (1.15)	0.96 (1.04)	1.13 (1.12)	1.16 (1.13)	1.12 (1.09)	10.9**
Board control (% of cases):						
VC always in control	100.0	0.0	5.9	5.0	0.0	81.5***
Founder always in control	0.0	3.5	0.0	0.0	93.8	81.4***
Outside directors pivotal	0.0	65.5	88.2	70.0	6.2	41.7***
Control state-contingent	0.0	31.0	5.9	25.0	0.0	13.2**
VC seats/CF, best case	1.13 (1.05)	0.67 (0.65)	1.21 (1.08)	1.32 (1.12)	1.23 (1.10)	40.2***
Founder seats/CF, best case	1.07 (0.92)	1.85 (1.28)	1.14 (0.85)	0.92 (0.92)	1.34 (1.12)	16.1***
VC liquidation rights (% of cases):						
Redemption rights	85.6	89.7	76.5	80.0	75.0	2.2
liq. claim > cum. investment	71.4	89.7	52.9	45.0	81.2	14.4***
Investment of VC funds:						
Investment commitment	20.1 (12.4)	7.09 (6.00)	2.95 (2.30)	3.21 (2.24)	3.77 (3.00)	25.1***
Sm % of funds released up front	54.6 (50.0)	73.3 (70.0)	97.6 (100)	97.6 (100.0)	90.6 (100.0)	29.9***
Number of VCs in syndicate	3.6 (3.0)	3.7 (4.0)	3.0 (2.0)	1.8 (2.0)	1.9 (2.0)	21.8***
Relative auto. conv. price:	3.14 (3.00)	5.30 (4.32)	5.13 (5.00)	3.8 (3.0)	3.30 (3.00)	5.7
Contingent contracting on (% of cases):						
Financial performance	28.6	34.5	23.5	0.0	37.5	9.4*
Non-financial performance	14.3	20.7	17.6	0.0	0.0	7.8*
Actions	35.7	24.1	5.9	0.0	0.0	14.9***
Default on redemp./dividend	28.6	41.4	11.8	40.0	18.8	6.3
Securities issues/valuations	50.0	17.2	29.4	20.0	37.5	6.3
Founder staying with firm	57.1	55.2	94.1	5.0	6.2	41.0***
% Pre-revenue ventures	64.3	58.6	62.5	35.0	25.0	8.7*

Table compares mean (median) frequencies and values of various contract provisions and firm characteristics between five clusters of portfolio companies. Data are taken from first VC financing rounds when available (89 cases) otherwise

Table 7: Complementarity and substitutability of contractual provisions.

6.7 Table 7: Complementarity and substitutability of contractual provisions

Read:

- The cluster analysis sorts contracts into five groups from maximum VC control to maximum founder control.
- Founder residual cash-flow rights are much lower in maximum-VC-control clusters and much higher in maximum-founder-control clusters.

- Voting control and board control move together.
- Strong VC control is associated with high use of contingent founder equity, staging, and liquidation rights.
- Pre-revenue ventures are more concentrated in high-VC-control clusters.

Interpretation: Table 7 is the main complementarity evidence. The rights are bundled in systematic ways rather than used one at a time.

Economic message: Real contracts solve agency problems with packages of instruments: cash flow, control, liquidation, staging, and contingencies are complements.

6.8 Table 8: Evolution of contracts over time

Read:

- Founder residual cash-flow rights decline across financing rounds.
- VC voting and board control generally increase in later rounds.
- Founder control becomes less common over time.
- Ex ante staged first rounds differ from non-staged first rounds: they involve larger commitments, less funding released up front, stronger VC control, and more contingencies.
- Syndicates become larger in later rounds.
- The average change between consecutive rounds shows a systematic shift of cash flow and control rights toward VCs.

Interpretation: Table 8 documents dynamic contracting. The terms of the relationship change as the firm raises more money, performance is revealed, and previous terms may be renegotiated.

Economic message: Venture capital contracting is a repeated financing process, not a one-shot security design problem.

Evolution of contracts over time						
Mean (median)	Round 1 Ex ante staging	Round 1 No ex ante staging	Round 2	Round 3	Round 4+	Average change between two consecutive rounds N = 90
	N = 34	N = 64	N = 50	N = 31	N = 34	
Founder residual cash flow rights, %:						
Total equity in best case	30.8 (30.3) ^[1]	44.0 (41.1) ^[1]	25.7 (21.1)	23.6 (21.3)	21.5 (16.0)	-10.2 (-7.7)***
Sensitivity, benchmarks	2.81 (0.00)	1.21 (0.00)	0.42 (0.00)	0.43 (0.00)	0.63 (0.00)	-0.38 (0.00)
Sensitivity, vesting	7.02 (3.21)	8.68 (0.00)	4.12 (0.00)	2.31 (0.00)	4.24 (0.00)	-3.69 (0.00)***
Voting control (% of cases):						
VC always in control	64.7 ^[1]	28.1 ^[1]	61.2	61.3	70.6	19.1 (0.00)***
VC always in control	8.82 ^[5]	28.1 ^[5]	8.2	3.2	0.0	-9.00 (0.00)***
Control state-contingent	17.6	28.1	14.0	9.7	8.8	-11.1 (0.00)**
VC votes/CF, best case	1.13 (1.09)	1.16 (1.13)	1.16 (1.14)	1.13 (1.10)	1.18 (1.14)	0.00 (0.01)
VC votes/CF, best case	0.98 (1.05) ^[5]	1.20 (0.99) ^[5]	1.09 (1.10)	1.13 (1.10)	1.09 (1.10)	0.01 (-0.01)
Board control (% of cases):						
VC always in control	23.3 ^[1]	4.7 ^[1]	27.3	34.5	54.6	11.0 (0.00)***
VC always in control	3.8	21.3	11.4	6.9	6.1	-8.75 (0.00)**
Outside directors pivotal	41.2 ^[5]	63.9 ^[5]	54.6	55.2	33.3	2.50 (0.00)
Control state-contingent	26.5 ^[5]	9.68 ^[5]	6.8	3.5	6.1	-4.94 (0.00)**
VC votes/CF, best case	1.00 (0.87) ^[5]	1.13 (1.00) ^[5]	0.88 (0.84)	0.91 (0.76)	0.97 (0.90)	-0.11 (-0.07)***
VC votes/CF, best case	1.41 (0.96)	1.20 (0.99)	1.90 (1.29)	2.17 (1.52)	2.49 (1.13)	0.49 (0.27)***
VC liquidation rights (% of cases):						
Redemption rights	84.4	80.3	71.4	71.0	88.2	-1.11 (0.00)
Liq. claim > cum. investment	72.7	67.2	62.5	69.0	87.9	4.71 (0.00)**

Table 8: evolution of contracts, part 1.

TABLE 8—Continued						
	Round 1 Ex ante staging	Round 1 No ex ante staging	Round 2	Round 3	Round 4+	Average change between two consecutive rounds N = 90
	N = 34	N = 64	N = 50	N = 31	N = 34	
<i>nd:</i>						
stment, Sm	12.5 (6.7) ^[1]	3.69 (3.0) ^[1]	6.3 (5.4)	7.4 (6.2)	7.2 (5.3)	2.01 (1.17)***
ed up front	50.1 (51.8) ^[1]	100.0 (100) ^[1]	88.4 (83.9)	90.0 (100)	95.6 (100)	0.75 (0.00)
in syndicate	3.0 (2.0)	2.6 (2.0)	4.8 (4.5)	6.4 (6.0)	6.6 (5.0)	1.59 (1.00)***
<i>price:</i>						
ting on	5.1 (4.1)	4.0 (3.0)	3.3 (2.8)	2.7 (2.5)	9.7 (2.2)	2.03 (−0.43)***
nance	41.2 ^[1]	12.5 ^[1]	14.0	6.5	17.6	−5.56 (0.00)
rformance	23.5 ^[1]	4.7 ^[1]	8.0	6.5	5.9	−2.22 (0.00)
	32.4 ^[1]	4.7 ^[1]	14.0	6.5	2.9	1.11 (0.00)
ip/dividend	35.3	26.6	30.0	12.9	32.3	−3.33 (0.00)***
'valuations	38.2 ^[10]	21.9 ^[10]	16.0	16.1	14.7	2.22 (0.00)
with firm	58.8	42.2	42.0	29.0	29.4	0.00 (0.00)
egotiation						
'cases):	—	—	30.0	29.0	26.5	—

financing round for 213 investments in 119 portfolio companies by 14 venture capital partnerships. Investments were 7 and 1999. For first financing rounds, we distinguish rounds where part of the committed financing is contingent on rounds where all funds are released upon the initial closing (no ex ante staging). Significant differences between the two stagings are indicated at the 1% ^[1], 5% ^[5], and 10% ^[10] levels, using a Mann-Whitney test. The last column shows the difference between two consecutive rounds using the sample of observations where this information is available. Significant results are indicated at the 1% ^{***}, 5% ^{**}, and 10% ^{*} levels, using a Wilcoxon test.

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Table 8: evolution of contracts, part 2.

7 Interpretive synthesis

7.1 Theory

- **Classical principal-agent theory.** Supported by pay-performance sensitivity in founder cash-flow rights, vesting, and performance benchmarks.
- **Incomplete contracts and control rights.** Supported by the central role of board and voting rights and their state-contingent transfer to VCs.
- **Debt and liquidation theories.** Supported by redemption, liquidation preferences, seniority, and downside control.
- **Hold-up and human-capital theories.** Supported by non-compete clauses, founder vesting, and contingencies on founder employment.
- **Dynamic theories of financing.** Supported by staging, automatic conversion, renegotiation, and evolution across financing rounds.

7.2 What theory still misses

The paper shows that real contracts simultaneously use many rights. Theory often studies a single instrument: debt versus equity, cash flow versus control, or staged financing. Real contracts combine all of these and make them contingent on many different states. The incremental theory challenge is to model complements among instruments and the evolution of rights through repeated rounds.

7.3 Practical interpretation

VC contracts allocate downside protection, upside participation, monitoring power, founder retention, and future financing options. The contract is not just a pricing agreement. It is an adaptive governance system for a high-uncertainty firm.

8 Short summary

- The paper studies 213 VC investments in 119 portfolio companies by 14 VC partnerships.
- It directly observes legal contracts rather than inferring contract terms from public securities data.
- VC contracts separate cash flow rights, board rights, voting rights, liquidation rights, redemption rights, and conversion rights.
- Rights are often contingent on performance, milestones, actions, default, securities offerings, and founder employment.
- Founder equity and control decline as performance worsens or later rounds occur.
- VC control and liquidation rights increase in bad states and later financing rounds.
- Contracts use incentives and control as complements, not substitutes.
- The evidence supports principal-agent, incomplete-contracting, debt, and hold-up theories, but shows contracts are richer than most models.
- The incremental contribution is a detailed empirical map from real VC contracts to financial-contracting theory.

9 Conclusion

9.1 Core conclusion

Kaplan and Strömberg show that real VC contracts look very much like financial-contracting theory in broad outline, but much more complex in detail. The contracts use separable and state-contingent allocations of cash flow, control, liquidation, conversion, vesting, and financing rights.

9.2 Economic intuition

Startups are uncertain, information is asymmetric, actions are hard to specify, founders' human capital is critical, and future financing is uncertain. Therefore, contracts must create incentives, allocate control in bad states, protect investors in liquidation, retain founders, and preserve financing options. No single security can do all of this alone, so VC contracts bundle many provisions.

9.3 Incremental contribution

The paper's enduring contribution is to make financial-contracting theory empirically concrete. It shows which theoretical mechanisms actually appear in contracts, which rights are separated, how they are made contingent, and how they evolve over financing rounds. It also provides stylized facts that motivate richer theories of multidimensional and dynamic contracting.

9.4 Takeaway

Real venture capital contracts are not simply convertible preferred stock. They are comprehensive governance systems that allocate value, control, downside protection, and future bargaining power across states of the world.